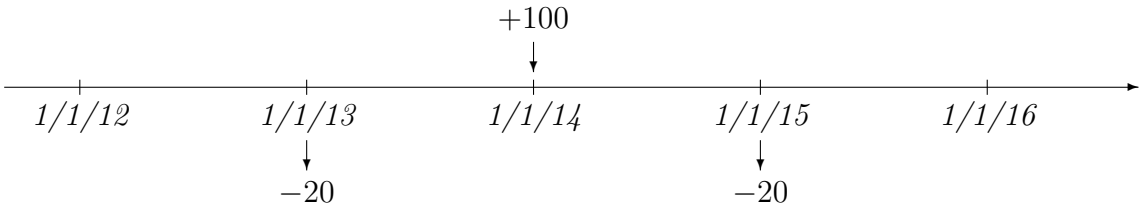
F78AB - Actuarial and Financial Mathematics B
Tutorial 2 – Measuring fund performance

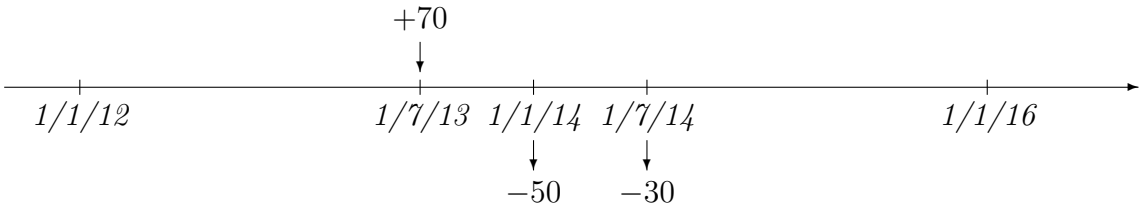
Question 1 A fund, which has value £100 on 1 January 2012, has the following cashflows in £:



The value of the assets in the fund on 1 January 2016 is £170.

1. Calculate the Money-Weighted Rate of Return (MWRR) during the calendar years 2012–2015.
2. If the fund value on 1 January 2014 was £100 just before the cash inflow on 1 Jan 2014 was received, calculate the Linked Internal Rate of Return (LIRR) during the calendar years 2012–2015, based on the two sub-periods calendar years 2012–2013 and 2014–2015.
3. If, in addition, the fund value was £110 on 1 January 2013 and £180 on 1 January 2015 (just before the cashflows on 1 January 2013 and 2015), calculate the Time-Weighted Rate of Return (TWRR) during the calendar years 2012–2015.

Question 2 A fund, which has value £100 on 1 January 2012, has the following cashflows in £:



The value of the assets in the fund on 1 January 2016 is £140.

1. Calculate the Money-Weighted Rate of Return (MWRR) during the calendar years 2012–2015.
2. If the fund value on 1 January 2013 was £105, and on 1 January 2015 was £130, calculate the Linked Internal Rate of Return (LIRR) during the calendar years 2012–2015, based on the three sub-periods 2012, 2013–2014 and 2015.
3. Suppose, in addition to the information in part 2, that you knew the following fund values:

Date	1 July 2013	1 January 2014	1 July 2014
Fund Value	£115	£195	£155

Calculate the Time-Weighted Rate of Return (TWRR) during the calendar years 2012–2015.

Question 3 [CT1, Apr 2010, Q10] A pension fund's assets were invested with two fund managers. On 1 January 2007, Manager A was given £120 000 and Manager B was given £100 000. A further £10 000 was invested with each manager on 1 January 2008 and again on 1 January 2009.

The values of the funds were:

Date	31 Dec 2007	31 Dec 2008	31 Dec 2009
Manager A	£130 000	£135 000	£180 000
Manager B	£140 000	£145 000	£150 000

1. Calculate the time-weighted rates of return earned by Manager A and Manager B over the period 1 January 2007 to 31 December 2009.

2. Show that the money-weighted rate of return earned by Manager A over the period 1 January 2007 to 31 December 2009 is approximately 9.4% per annum.

3. Explain, without performing further calculations, whether the money-weighted rate of return earned by Manager B over the period 1 January 2007 to 31 December 2009 was higher than, lower than or equal to that earned by Manager A.

4. Discuss the relative performance of the two fund managers.

Question 4 [CT1, Sep 2011, Q4] A pension fund makes the following investments in £m:

Date	1 Jan 2009	1 Jul 2009	1 Jan 2010
Investment amounts (£m)	1.5	6.0	4.0

The rates of return earned on money invested in the fund were as follows:

Time period	Rate of return (i.e. not annualised)
1 Jan 2009 to 30 Jun 2009	1%
1 Jul 2009 to 31 Dec 2009	2%
1 Jan 2010 to 31 Dec 2010	5%

Assume that 1 January to 30 June and 1 July to 31 December are precise half-year periods.

1. Calculate the time-weighted rate of return per annum effective over the two years from 1 January 2009 to 31 December 2010.

2. Calculate the money-weighted rate of return per annum effective over the two years from 1 January 2009 to 31 December 2010.

Solutions: Q1.1. 1.878%; Q1.2. 3.706%; Q1.3. 3.975%; Q2.1. 9.969%; Q2.2. 9.652%; Q2.3. 9.758%; Q3.1 A: 9.050% p.a. B: 9.409% p.a.; Q3.3. Lower; Q4.1. 4.005% p.a.; Q4.2 4.63% p.a.

F78AB - Actuarial and Financial Mathematics B
Solutions to Tutorial 2 – Measuring fund performance

Solution 1

1. The equation of value is:

$$f(i) := 100\,(1+i)^4 - 20\,(1+i)^3 + 100\,(1+i)^2 - 20\,(1+i) - 170 = 0.$$

Denote the solution to the equation of value by $i^\star$. After trial and error, $f(0.01) = -4.736$ and $f(0.02) = +0.659$. Hence by linear interpolation between these two points,

$$i^\star \approx 0.01 + 0.01\,\frac{4.736}{4.736 + 0.659} = 0.01878.$$

2. Denote the first return in 2012–2013 by $i_1^\star$ and the second return in 2014–2015 by $i_2^\star$.

For 2012–2013, the equation of value is: $f_1(i) := 100\,(1+i)^2 - 20\,(1+i) - 100 = 0$.

As this is a quadratic equation, we solve it exactly to find that the only positive root is $1 + i_1^\star = 1.10499$.

Do not linearly interpolate when you can determine the exact answer.

For 2014–2015, the equation of value is: $f_2(i) := 200\,(1+i)^2 - 20\,(1+i) - 170 = 0$. Again, we solve it exactly to find that the only positive root is $1 + i_2^\star = 0.97331$.

The (annual) LIRR is denoted LIRR. Hence, as there are two years in 2012–2013 and two years in 2014–2015,

$$(1 + \text{LIRR})^4 = (1 + i_1^\star)^2\,(1 + i_2^\star)^2 = 1.10499^2 \times 0.97331^2 \quad \Rightarrow \quad \text{LIRR} = 0.03706.$$

3. We calculate the growth factors for each period:

$$\begin{aligned}(1 + \text{TWRR})^4 &= \frac{110}{100}\,\frac{100}{(110-20)}\,\frac{180}{(100+100)}\,\frac{170}{(180-20)} = 1.16875 \\ \Rightarrow \quad \text{TWRR} &= 0.03975.\end{aligned}$$

Solution 2

1. Work in time units of half a year. The equation of value is:

$$f(i) := 100\,(1+i)^8 + 70\,(1+i)^5 - 50\,(1+i)^4 - 30\,(1+i)^3 - 140 = 0.$$

Denote the solution to the equation of value by $i^\star$. After trial and error, $f(0.04) = -10.216$ and $f(0.05) = +1.581$. Hence by linear interpolation between these two points,

$$i^\star \approx 0.04 + 0.01\,\frac{10.216}{10.216 + 1.581} = 0.04866.$$

Annualising this half-yearly interest rate, we find that the MWRR is 0.09969 per annum.

2. Denote the first return in 2012 by $i_1^\star$, the second return in 2013–2014 by $i_2^\star$ and the third return in 2015 by $i_3^\star$.

   For 2012, $i_1^\star = 105/100 - 1 = 0.05$ per annum.

   For 2013–2014, in time units of half a year, the equation of value is:

$$f_2(j) := 105(1+j)^4 + 70\,(1+j)^3 - 50\,(1+j)^2 - 30\,(1+j) - 130 = 0.$$

   Denote the solution to the equation of value by $j^\star$. After trial and error, $f_2(0.06) = -2.049$ and $f_2(0.07) = +4.042$. Hence by linear interpolation between these two points,

$$j^\star \approx 0.06 + 0.01\,\frac{2.049}{2.049 + 4.042} = 0.06336.$$

   Annualising the figure, $i_2^\star = 0.1307$ per annum.

   For 2015, $i_3^\star = \dfrac{140}{130} - 1 = 0.07692$.

   Hence $(1 + \text{LIRR})^4 = 1.05 \times 1.1307^2 \times 1.07692 \qquad \Rightarrow \qquad \text{LIRR } = 0.09652.$

3. We find growth factors for each period:

$$\begin{aligned}(1 + \text{TWRR})^4 &= \frac{115}{100}\,\frac{195}{(115+70)}\,\frac{155}{(195-50)}\,\frac{140}{(155-30O)} \\ \Rightarrow \qquad \text{TWRR } &= 0.09758.\end{aligned}$$

   Note: It could use the information in part 2 to find the TWRR. However, since no cashflows occur on 1 January 2013 or 1 January 2015, the growth factors would just cancel out.

Solution 3

1. The values of the funds before and after the cash injections are:

	Manager A		Manager B	
Date	Before	After	Before	After
1 Jan 2007	120 000		100 000	
31 Dec 2007	130 000	140 000	140 000	150 000
31 Dec 2008	135 000	145 000	145 000	155 000
31 Dec 2009	180 000		150 000	

   Thus, TWRR for Manager A is given by:

$$(1 + i_A)^3 = \frac{130}{120} \times \frac{135}{140} \times \frac{180}{145} \qquad \Rightarrow i_A = 0.09050$$

   and the TWRR for Manager B is given by:

$$(1 + i_B)^3 = \frac{140}{100} \times \frac{145}{150} \times \frac{150}{155} \qquad \Rightarrow i_B = 0.09409$$

2. MWRR for Manager A is given by:

$$120 \times (1+i)^3 + 10 \times (1+i)^2 + 10 \times (1+i) = 180.$$

   Then, putting $i = 0.094$ gives the left-hand side of the above equation equals 180.03, which is close enough to 180.

3. Both funds increased by 50% over the three-year period and received the same cashflows at the same times.

   Since the initial amount invested in Manager B's fund was lower, the cash inflows received represent a larger proportion of Manager B's fund. Since Manager B's returns were negative for both the 2nd and 3rd years, whereas Manager A had only a negative return in the 2nd year (which was roughly of the same magnitude as Manager B's 2nd year return) the money-weighted return earned by Manager B over the period will be lower.

4. The money-weighted rate of return is higher for Manager A, whilst the time-weighted return is higher for Manager B. When comparing the performance of investment managers, the time-weighted rate of return is generally better because it ignores the effects of cash inflows or outflows being made which are beyond the manager's control.

   In this case, Manager A's best performance is in the final year, when her fund was at its largest, whilst Manager B's best performance was in the first year, where his fund was at its lowest.

   Overall, it may be argued that Manager B has performed slightly better than Manager A since Manager B achieved the higher time-weighted return.

Solution 4

1. - Either, calculate directly from the rates of return:
     The TWRR is i such that

     $$(1+i)^2 = 1.01 \times 1.02 \times 1.05 = 1.08171 \qquad \Rightarrow \qquad i = 4.005\%.$$

   - Or, calculate the fund values just before each time a cashflow occurs.
     The fund value on 30 June 2009 will be $1.5 \times 1.01 = 1.515$.
     The fund value on 31 December 2009 will be $(1.515 + 6) \times 1.02 = 7.6653$.
     The fund value on 31 December 2010 will be $(7.6653 + 4) \times 1.05 = 12.2486$.
     The TWRR is i such that

     $$(1+i)^2 = \frac{1.515}{1.5} \times \frac{7.6653}{7.515} \times \frac{12.2486}{11.6653} = 1.0817 \qquad \Rightarrow \qquad i = 4.005\%.$$

2. The equation of value is

   $$1.5(1+i)^2 + 6.0(1+i)^{1.5} + 4(1+i) - 12.2486$$

   Trial-and-error results in: $f(4\%) = -0.1026$, $f(4.5\%) = -0.0210$ and $f(5\%) = 0.0607$.

   Interpolate between $i = 4.5\%$ and $i = 5\%$ to find

   $$i = 0.045 + (0.05 - 0.045) \times \frac{0.0210}{0.0210 + 0.0607} = 0.0463.$$